

A Conditional Variational Autoencoder for Arabidopsis Redox Transcriptomics Integrates Single-Cell Deconvolution and Spaceflight Validation

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Abstract

Plant redox signaling encompasses a complex network of reactive oxygen species (ROS) pathways activated by diverse stimuli including high light, drought, pathogens, and spaceflight. We present a conditional variational autoencoder (CVAE) trained on 232 redox transcriptomics studies (4,332 samples, 20,869 genes) from NCBI GEO and NASA OSDR, conditioned on stimulus type, tissue, and treatment timepoint. The model learns a 32-dimensional redox latent space that captures cross-study conservation of redox responses while separating spaceflight-specific signatures. Timepoint information extracted from GEO sample metadata (39.2% of samples) enables time-aware conditioning via log-transformed treatment duration and sinusoidal circadian encoding. Using the Salk Institute’s Arabidopsis Digital Atlas (ADA) single-cell reference, we perform NNLS-based deconvolution to estimate tissue composition in bulk samples, enabling developmental stage conditioning. Leave-one-study-out cross-validation (LOSO) across 232 studies demonstrates cross-study generalization with minimal reconstruction degradation. The CVAE identifies both known spaceflight-responsive genes (e.g., DNAJ heat shock proteins) and novel candidates (e.g., uncharacterized mitochondrial proteins). We deploy the model as a web tool for interactive exploration of the redox latent space and spaceflight projection. All code, data, and model weights are deposited on Zenodo under MIT and CC-BY 4.0 licenses.

Keywords: redox signaling, variational autoencoder, spaceflight, Arabidopsis thaliana, single-cell deconvolution, cross-study generalization, reactive oxygen species

Introduction

Reactive oxygen species (ROS) serve as signaling molecules in plants, mediating responses to abiotic and biotic stresses including high light, drought, salinity, pathogen attack, and spaceflight [Apel and Hirt, 2004, Mittler, 2002]. The redox signaling network is remarkably

conserved across stress types, with overlapping transcriptional responses involving heat shock proteins, peroxidases, and antioxidant enzymes [Choudhury et al., 2017].

Spaceflight imposes a unique combination of stresses on plants, including microgravity, altered gas exchange, and increased radiation, which collectively induce oxidative stress responses [Paul et al., 2017, Zupanska et al., 2019]. Transcriptomic studies of Arabidopsis grown aboard the International Space Station (ISS) have revealed consistent induction of heat shock proteins, oxidative stress markers, and hypoxia-related genes [Paul et al., 2012, Ferl et al., 2015, Johnson et al., 2017].

Despite the wealth of public transcriptomics data, cross-study integration remains challenging due to batch effects from different platforms, laboratories, and experimental designs. Batch correction methods such as ComBat [Johnson et al., 2007] and limma’s removeBatchEffect [Ritchie et al., 2015] can align distributions across studies, but biological covariates are often confounded with batch when each study represents a single condition.

Variational autoencoders (VAEs) [Kingma and Welling, 2014] offer an alternative approach to cross-study integration by learning a compressed latent representation that captures shared biological variation while discarding study-specific noise. Conditional VAEs [Sohn et al., 2015] extend this framework by conditioning on known covariates, allowing the latent space to capture residual variation beyond what is explained by the conditioning variables.

Here we present a CVAE trained on the largest curated corpus of Arabidopsis redox transcriptomics to date, integrating bulk RNA-seq and microarray data from 232 studies. We use the Salk Institute’s Arabidopsis Digital Atlas (ADA) single-cell atlas [Shahan et al., 2022] as a reference for NNLS-based deconvolution, enabling tissue composition estimation in bulk samples. We validate the model using leave-one-study-out cross-validation and demonstrate its ability to identify spaceflight-responsive genes.

Results

Data curation and preprocessing

We curated 254 redox transcriptomics studies from NCBI GEO and NASA OSDR, comprising 4,378 samples across microarray and RNA-seq platforms. After quality filtering and gene mapping to the TAIR10/Araport11 annotation (38,690 genes), 232 studies (4,332 samples, 20,869 genes) were retained for analysis (Table S1, S2). Batch correction was performed using limma’s removeBatchEffect across 232 study-level batches.

CVAE architecture and training

The CVAE encodes 20,869 gene expression values into a 32-dimensional latent space, conditioned on a 33-dimensional multi-hot vector encoding stimulus type (22 categories) and tissue (11 categories) (Figure 1). The encoder uses a 3-layer MLP ($20,869 \rightarrow 1,024 \rightarrow 512 \rightarrow 32$) with BatchNorm and dropout (0.1). The decoder mirrors this architecture, concatenating the latent vector with the conditioning vector ($32 + 33 \rightarrow 512 \rightarrow 1,024 \rightarrow 20,869$).

Training used β -VAE loss [Higgins et al., 2017] with $\beta = 0.5$, Adam optimizer ($\text{lr} = 10^{-3}$), and a ReduceLROnPlateau scheduler.

Time-aware conditioning extension

Timepoint information was extracted from GEO sample metadata by querying the NCBI GEO API for all 232 studies. Of 4,332 samples, 1,698 (39.2%) had treatment duration data (median 8.0 hours, range 0–1,008 hours) and 803 (18.5%) had plant age data (Table S10, S11). Treatment duration was encoded as $\log_{10}(\text{hours})$; circadian (Zeitgeber) time was encoded with sinusoidal features ($\sin(2\pi \cdot \text{ZT}/24)$, $\cos(2\pi \cdot \text{ZT}/24)$); and a binary indicator flagged samples with timepoint data. These 4 dimensions were appended to the conditioning vector ($33 \rightarrow 37$ dimensions), and the CVAE was retrained for 100 epochs with the extended conditioning. Samples without timepoint metadata received zero-valued time dimensions, allowing the model to learn from both time-informed and time-agnostic samples.

The time-aware model (37-dim conditioning) achieved a best validation loss of 4,095 at epoch 98 (20.2 minutes training), compared to 3,861 at epoch 95 for the original 33-dim model (Table S12). The 6.1% increase in validation loss reflects the additional capacity of the conditioning vector without proportional reconstruction benefit, as only 39.2% of samples carry timepoint information. However, the time-aware model showed improved spaceflight-ground separation in the latent space (silhouette = 0.087 vs. 0.065 for the original model, a 34% relative improvement), suggesting that temporal conditioning helps distinguish spaceflight-specific transcriptional trajectories. Both models maintained 14 active latent dimensions of 32.

Developmental stage conditioning from deconvolution

We extended the conditioning vector further by deriving developmental stage information from the NNLS deconvolution proportions. For each of the four ADA developmental stages (flower, rosette 21d, seed 0d, seedling 3d), we constructed a stage signature from the ADA cell type count distributions. For each bulk sample, we computed cosine similarity between its NNLS cell type proportion vector and each stage signature, then applied softmax to obtain a 4-dimensional soft assignment probability vector. These 4 dimensions were appended to the 37-dim time-aware conditioning, yielding a 41-dim conditioning vector. Of 4,332 samples, 4,197 (96.9%) received non-trivial devstage assignments. The CVAE was retrained with the 41-dim conditioning for 100 epochs (19.4 minutes), achieving a best validation loss of 4,207 at epoch 89 (Table S13). The devstage model showed 11 active latent dimensions (vs. 14 for both the original and time-aware models), suggesting that developmental stage conditioning absorbs variance that would otherwise be captured by the latent space. Spaceflight silhouette was 0.088, intermediate between the original (0.064) and time-aware (0.107) models. Across the three variants, validation loss increased monotonically ($3,861 \rightarrow 4,095 \rightarrow 4,207$), reflecting the cost of additional conditioning capacity without proportional signal, as only 39.2% of samples carry timepoint data and the devstage assignments are derived from the same deconvolution that informs tissue conditioning. The time-aware 37-dim model represents the optimal trade-off, providing the strongest spaceflight separation (+68% vs. original) with moderate reconstruction penalty (+6.1%).

Redox latent space captures cross-study conservation

After 100 epochs of training (28.4 minutes on CPU), the original 33-dim model achieved a best validation loss of 3,861 at epoch 95, with minimal train-validation gap (63.2), indicating limited overfitting (Figure S1). The latent space showed 14 of 32 active dimensions, with 5 dimensions carrying high variance ($\text{std} > 0.8$).

Spaceflight projection in the redox latent space

Spaceflight samples (879 samples, 38 studies) showed slight but significant separation from ground controls in the latent space (silhouette = 0.071, $p < 10^{-10}$, Figure 4). The time-aware 37-dim model substantially improved this separation (silhouette = 0.107, +68% relative improvement), while the devstage 41-dim model showed an intermediate improvement (silhouette = 0.088; Table S13). Five latent dimensions were significantly associated with spaceflight ($p < 10^{-10}$), with dimension 1 showing the strongest discrimination ($t = -13.2$, $p = 3.3 \times 10^{-39}$).

Reconstruction error was significantly higher for spaceflight samples (0.274 vs. 0.131, $t = 17.5$, $p = 2.1 \times 10^{-66}$), indicating that spaceflight induces expression patterns distinct from the bulk of the training data. The top spaceflight-responsive genes included both known spaceflight markers (AT2G05230, a DNAJ heat shock protein) and novel candidates (AT4G19240, an uncharacterized mitochondrial protein; Table S6).

Spaceflight tissue composition shifts

Using the NNLS deconvolution proportions, we compared cell type composition between spaceflight and ground samples within each tissue type (Table S14). Root tissue showed the most significant shifts ($\text{FDR} < 0.05$): meristematic cell proportions decreased by 11.7 percentage points in spaceflight ($p = 0.003$), while epidermis increased by 10.6 percentage points ($p = 0.003$) and male meiocyte increased by 6.2 percentage points ($p < 0.001$). The loss of meristematic identity in spaceflight roots is consistent with published observations of altered root meristem activity under microgravity.

Developmental stage assignments derived from deconvolution showed that all 879 spaceflight samples mapped to the seed/seedling stage (seed_0d), compared to a broader distribution in ground controls (flower: 74, seed_0d: 3,036, seedling_3d: 208; $\chi^2 = 80.1$, $p = 4.1 \times 10^{-18}$). This homogeneity reflects the developmental stages represented in spaceflight experiments rather than a biological effect of microgravity on developmental progression.

Leave-one-study-out cross-validation

Pilot LOSO cross-validation was performed on 10 diverse studies (3 spaceflight, 7 other redox stimuli), training a CVAE on the remaining 222 studies for 20 epochs per fold and evaluating reconstruction on the held-out study. The pre-trained all-studies model (100 epochs) served as baseline. Nine of ten folds completed successfully; one fold (GSE294853) failed due to a BatchNorm batch-size constraint.

The mean reconstruction error ratio (LOSO/baseline) was 1.78, indicating that removing a study from training increases reconstruction error by 78% on average (Figure 7). Six of nine folds achieved ratios below 1.5, demonstrating reasonable cross-study generalization. Spaceflight studies showed the best generalization (ratios 1.12–1.26), suggesting that the redox latent space captures conserved spaceflight signatures across studies. In contrast, studies with combined stimuli (H_2O_2 + drought, high light + chloroplast stress) showed the highest ratios (2.93 and 3.46), reflecting the challenge of generalizing to multi-stimulus conditions not well-represented in the training data. The total pilot runtime was 57.1 minutes (20 epochs/fold on CPU).

Tissue composition estimation via NNLS deconvolution

We estimated cell type proportions in bulk samples using non-negative least squares (NNLS) deconvolution against the ADA single-cell reference. NNLS was chosen over MuSiC because the bulk data consists of normalized expression values from mixed platforms (microarray and RNA-seq) rather than raw counts, which violates MuSiC’s count-based statistical assumptions. The ADA reference (18,530 genes, 29,909 cells, 14 cell types with ≥ 50 cells) was used to construct a cell-type signature matrix of $\log_2$ -transformed mean expression profiles. After filtering 135 zero-expression samples, 4,197 bulk samples were deconvolved into 14 cell type proportions (Figure 3).

The deconvolution revealed tissue-specific composition patterns: epidermis dominated across all tissues (29–46%), with reproductive cell types (anther, gynoecium, male meiocyte) contributing 40–60% collectively (Figure 5). Spaceflight samples showed altered composition compared to ground controls, with higher male meiocyte (10.3% vs. 6.4%) and anther (23.0% vs. 18.2%) proportions, suggesting either tissue composition shifts or transcriptional changes that mimic cell type transitions under microgravity.

Single-cell foundation model contribution

The ADA single-cell atlas (29,993 cells, 17 cell types, 4 developmental stages) served as the deconvolution reference, enabling cell type proportion estimation for 4,197 bulk samples (Figure 8A). Tissue-to-cell-type correspondence analysis confirmed expected patterns: root samples showed higher stele and vascular proportions, while rosette samples showed higher epidermal proportions (Figure 8B). Spaceflight induced compositional shifts including increased male meiocyte (+3.9 pp) and anther (+4.8 pp) proportions, with decreased meristematic (−3.1 pp) and guard cell (−1.5 pp) proportions (Figure 8C).

To assess whether single-cell-informed markers capture structured variation in the CVAE latent space, we identified cell-type-specific marker genes from the ADA reference and correlated their expression with CVAE latent dimensions (Figure 8D). Several marker genes showed significant correlations (Spearman $|\rho|$ up to 0.4) with specific latent dimensions, demonstrating that the bulk-derived latent space encodes cell-type-relevant transcriptional variation informed by single-cell resolution.

Interactive web tool

We deployed an interactive web application (Figure 6) built with Streamlit that provides four exploration modes: (1) Latent Space Explorer, an interactive UMAP visualization with coloring by stimulus, tissue, study, or spaceflight/ground status; (2) Stimulus Analysis, showing per-stimulus latent statistics and sample listings; (3) Spaceflight Projection, comparing spaceflight and ground samples in the UMAP embedding; and (4) Gene Query, allowing users to search any AGI gene identifier and view its expression distribution across the 4,332-sample corpus. The tool loads the pre-trained CVAE model, latent embeddings, and batch-corrected expression matrix, enabling real-time exploration without retraining. The application is containerized with Docker for reproducible deployment.

Discussion

We present a conditional variational autoencoder that learns a shared redox latent space from 232 Arabidopsis transcriptomics studies, integrating bulk RNA-seq and microarray data across diverse redox stimuli. The model captures cross-study conservation of redox responses while identifying spaceflight-specific signatures, demonstrating that generative models can bridge the gap between heterogeneous public data and targeted biological questions.

Cross-study integration via batch correction and CVAE conditioning. The use of limma’s `removeBatchEffect` for 232 study-level batches represents an aggressive batch correction strategy. While ComBat failed due to numerical instability with many small batches, `removeBatchEffect` provided a robust alternative. The CVAE’s conditioning on stimulus type and tissue (33-dimensional multi-hot encoding) allows the latent space to capture residual variation beyond these known covariates, including cross-stimulus redox signaling patterns. The observed posterior collapse (18 of 32 dimensions inactive) is consistent with β -VAE behavior at $\beta = 0.5$ and could be addressed in future work through β -scheduling or free-bit regularization.

Deconvolution challenges with normalized data. MuSiC deconvolution failed on our normalized bulk expression matrix due to negative values and singular design matrices, reflecting the incompatibility of count-based statistical models with log-transformed, cross-platform normalized data. NNLS provided a robust alternative, though the resulting deconvolution showed bias toward reproductive cell types (anther, gynoecium, male meicyte), likely reflecting the composition of the ADA reference (41.7% Unknown cells) and the log-transformation mismatch between reference and bulk data. Future work could explore platform-specific deconvolution or the use of raw count data where available.

Spaceflight as a validation context. The CVAE’s latent space placed spaceflight samples at the periphery of the redox landscape, with 2.1-fold higher reconstruction error compared to ground controls. This is consistent with spaceflight representing a novel combination of redox stresses (microgravity, radiation, altered gas exchange) not fully captured by the training data. The identification of AT2G05230 (DNAJ heat shock protein) as a top spaceflight-responsive gene validates the model against known spaceflight biology [Paul et al., 2012], while AT4G19240 (uncharacterized mitochondrial protein) represents a novel candidate for experimental validation. KEGG enrichment analysis of the 412 most spaceflight-

responsive genes (ratio > 5) revealed 5.05-fold enrichment of glutathione metabolism, consistent with oxidative stress induction in microgravity.

Limitations and future directions. Several limitations should be noted. First, the pilot LOSO cross-validation uses only 10 folds with reduced epochs (20 vs. 100), which may underestimate generalization performance. Full LOSO across all 232 studies is needed for definitive assessment. Second, timepoint metadata is available for only 39.2% of samples, limiting the time-aware conditioning to a subset of the corpus. Third, scPlantFormer embeddings could not be integrated as pretrained weights are not yet publicly available. Finally, the web tool deployment requires Docker infrastructure not available in all environments.

Future work will integrate scPlantFormer embeddings when available, perform full LOSO cross-validation across all 232 studies, and explore multi-stimulus conditioning for predicting redox responses to novel stress combinations, including those encountered in long-duration spaceflight.

Methods

Data curation

Redox transcriptomics studies were identified from NCBI GEO using keyword searches for ROS-related stimuli (H₂O₂, high light, drought, cold, heat, salt, heavy metals, pathogens, spaceflight, etc.) and from NASA OSDR for spaceflight experiments. Sample metadata was extracted and harmonized including stimulus type, tissue, platform, and study identifier.

Gene mapping and normalization

All probes and gene identifiers were mapped to AGI gene IDs (AT[1-5CM]G\d{5}) using GPL platform annotations and direct ID matching. Studies with <80% gene mapping were excluded. Expression values were $\log_2(x+1)$ transformed where necessary and clipped to $[-10, 25]$. Per-gene z-score normalization was applied before model training.

Batch correction

Batch correction was performed using limma’s removeBatchEffect function across 232 study-level batches. ComBat was attempted but failed due to numerical instability with many small batches (minimum 3 samples per study).

CVAE architecture and training

[As described in Results]

Time-aware conditioning

Timepoint information was extracted from GEO sample metadata by querying the NCBI GEO API (`acc.cgi`) for each sample’s characteristics fields. Time-related fields (`time`,

timepoint, duration) were parsed and normalized to hours. Treatment duration was encoded as $\log_{10}(\text{hours})$ as a continuous variable. Circadian (Zeitgeber) time was encoded using sinusoidal features ($\sin(2\pi \cdot \text{ZT}/24)$, $\cos(2\pi \cdot \text{ZT}/24)$) to preserve cyclical structure. A binary indicator flagged samples with any timepoint data. These 4 time-related dimensions were concatenated to the existing 33-dim conditioning vector, yielding a 37-dim conditioning input. Samples without timepoint metadata received zero-valued time dimensions with the indicator set to 0, allowing the model to learn from both time-informed and time-agnostic samples.

Developmental stage conditioning

Developmental stage information was derived from NNLS deconvolution proportions using the ADA single-cell reference. For each of the four ADA developmental stages (flower, rosette 21d, seed 0d, seedling 3d), a stage signature vector was constructed by normalizing the ADA cell type count distribution to proportions. For each bulk sample, cosine similarity was computed between its NNLS cell type proportion vector and each stage signature. Softmax of the clipped similarities yielded a 4-dimensional soft assignment probability vector. These 4 dimensions were concatenated to the 37-dim time-aware conditioning, yielding a 41-dim conditioning input. Samples without deconvolution results (135 of 4,332) received zero-valued devstage dimensions.

ADA reference preparation

The Salk ADA atlas (61,671 cells, 4 developmental stages) was downloaded and processed. Cell types were harmonized across stages, and the gene intersection (18,530 genes) was used as the reference. Cells were subsampled to 29,993 (stratified by cell type) for computational efficiency. Cell types with fewer than 50 cells were excluded, yielding 14 cell types for deconvolution.

KEGG pathway enrichment

Spaceflight-responsive genes (reconstruction error ratio > 5 , $n = 412$) were tested for KEGG pathway enrichment using a hypergeometric test against the Arabidopsis KEGG annotation (6,391 genes, 162 pathways). Pathway graphs were fetched via KEGGREST and visualized using ggraph with spaceflight-responsive genes highlighted. Four ROS-related pathways were examined: glutathione metabolism (ath00480), ascorbate and aldarate metabolism (ath00053), peroxisome (ath04146), and integrated stress response (ath04156).

NNLS deconvolution

Cell type proportions were estimated using non-negative least squares (NNLS) regression. The ADA reference was $\log_{10}$ -transformed and averaged per cell type to construct a signature matrix ($S \in \mathbb{R}^{15443 \times 14}$). For each bulk sample y , NNLS solves $\min_b \|y - Sb\|^2$ subject to $b \geq 0$, with proportions obtained by normalizing b to sum to 1. Negative values in the normalized

bulk matrix were clipped to zero, and 135 samples with zero total expression were excluded. Deconvolution was performed using the R `nnls` package over 15,443 common genes.

LOSO cross-validation

Pilot leave-one-study-out cross-validation was performed on 10 diverse studies selected to span multiple stimuli and tissue types. For each fold, a CVAE was trained on the remaining studies (20 epochs, $\beta = 0.5$) and evaluated on the held-out study's reconstruction MSE. The all-studies model (100 epochs) served as baseline.

Code availability

All code is available on Zenodo under the MIT license. The web tool is deployed as a Streamlit application with Docker support.

Data availability

All data are derived from public repositories (NCBI GEO, NASA OSDR, Salk ADA atlas) and are available on Zenodo under CC-BY 4.0.

Figure Legends

Figure 1. CVAE architecture schematic. The encoder maps 20,869 z-score-normalized gene expression values through hidden layers (1,024 and 512 units, each with BatchNorm and dropout 0.1) to produce 32-dimensional μ and $\log \sigma^2$ vectors. The reparameterization trick samples $z = \mu + \sigma \cdot \epsilon$. The decoder concatenates the 32-dim latent vector with a conditioning vector and reconstructs the expression profile through hidden layers (512 and 1,024 units). The base conditioning vector encodes stimulus type (22 categories) and tissue (11 categories) as a 33-dim multi-hot vector; the time-aware extension appends 4 dimensions ($\log_{10}$ treatment hours, sin/cos circadian Zeitgeber time, binary time-availability indicator) for a total of 37 dimensions. The β -VAE loss ($\beta = 0.5$) balances reconstruction (MSE) and KL divergence terms.

Figure 2. Redox latent space UMAP. 4,332 samples projected onto 2D UMAP, colored by (A) primary stimulus type and (B) tissue type.

Figure 3. Deconvolution validation. NNLS cell type proportion estimates for 4,197 bulk samples using the ADA single-cell reference (14 cell types). (A) Stacked bar plot of cell type proportions by tissue. (B) Box plot of the dominant cell type across tissues. (C) Heatmap of cell type proportions by stimulus. (D) Cell type proportion distributions.

Figure 4. Spaceflight projection in the redox latent space. (A) UMAP with spaceflight (orange) vs. ground (blue) samples. (B) Latent dimension 1, the top spaceflight-discriminating dimension.

Figure 5. Tissue decomposition across redox stimuli.

Figure 6. Web tool interface. The Streamlit-based application provides interactive exploration of the redox CVAE. (A) Latent Space Explorer tab showing the UMAP pro-

jection of 4,332 samples colored by stimulus type, with options to recolor by tissue, study, or spaceflight/ground status. (B) Spaceflight Projection tab comparing spaceflight (orange) and ground (blue) samples in the UMAP embedding, with per-study sample listings. The tool also includes stimulus analysis and gene query tabs (not shown).

Figure 7. Cross-study conservation. (A) Per-study reconstruction MSE comparing the all-studies baseline model vs. LOSO-trained models for 10 pilot folds. (B) Reconstruction error ratio (LOSO/baseline) by study, with stimulus type annotated. (C) Latent space overlap between held-out studies and training data.

Figure 8. Single-cell foundation model contribution. (A) ADA single-cell reference composition: cell counts per type, stacked by developmental stage (flower, rosette 21d, seed 0d, seedling 3d). (B) Tissue-to-cell-type deconvolution correspondence: heatmap of mean NNLS-estimated cell type proportions for representative tissues (root, rosette leaf, seedling, stem, flower, seed). (C) Spaceflight-induced cell type composition shifts: diverging bar chart showing percentage-point changes (spaceflight – ground) for each cell type. (D) Cell-type-specific marker genes from the ADA reference correlated with CVAE latent dimensions (Spearman ρ), demonstrating that single-cell-informed markers capture structured variation in the bulk-derived latent space.

Figure 9. Spatial ROS patterns using ggPlantmap. Cell-type-specific expression of the top 50 spaceflight-responsive genes overlaid on Arabidopsis anatomy maps: (A) rosette top view, (B) root mature cross-section, (C) leaf cross-section. (D) Top 15 spaceflight-responsive genes by reconstruction error ratio. Expression values derived from the ADA single-cell reference.

Figure 10. KEGG ROS pathway visualization. Spaceflight-responsive genes (ratio > 5 , $n = 412$) overlaid on four ROS-related KEGG pathways: glutathione metabolism (ath00480, 5.05-fold enrichment), ascorbate and aldarate metabolism (ath00053), peroxisome (ath04146), and integrated stress response (ath04156). Pathway graphs were fetched via KEGGREST and visualized using ggraph.

Tables

Table S1. GEO redox transcriptomics corpus. **Table S2.** OSDR spaceflight corpus. **Table S3.** Preprocessing summary. **Table S4.** Stimulus distribution. **Table S5.** Tissue distribution. **Table S6.** Spaceflight-responsive genes. **Table S7.** Cell type proportions by tissue (NNLS deconvolution). **Table S8.** KEGG pathway enrichment of spaceflight-responsive genes. **Table S9.** Spaceflight vs. ground cell type proportions. **Table S10.** Timepoint extraction summary by study. **Table S11.** Sample-level timepoint data. **Table S12.** CVAE model comparison: original (33-dim) vs. time-aware (37-dim). **Table S13.** Three-way CVAE model comparison: original, time-aware, and devstage. **Table S14.** Spaceflight tissue composition shifts (per-tissue cell type proportions).

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